Bench exercise: mark out an okkake, and plan its cuts

Japanese Joinery by Hand · Episode 12. Mark the lower piece of an okkake daisen-tsugi 追掛け大栓継ぎ on a block of softwood in the training course's order. Then write the cutting sequence on it in pencil before you touch a saw. Cutting is optional: do it with a teacher if you can. About 60–90 minutes for the marking.

> **Safety.** If you go on to cut: clamp the work, keep both hands behind the edge, and stop saw cuts at the line. Sharp-tool technique, and how to use these tools safely, is best learned from a teacher, at a bench, in person.

1 · Prepare

1. A planed softwood block about 60 × 90 mm and 500 mm long (half the episode's 120 × 180). Splice length: 180 mm (twice the depth, like the model). Step and tongue: about 8 mm.
2. Strike a center line on the top face, as in Episode 1.

2 · Mark (the course's order)

1. Top face: the end line, then the housing for the other piece's tongue.
2. The shoulder lines and the line along the beam; then the other shoulder, with marks to locate the slope.
3. The slope line on the top face and on one side face, with that face's shoulder lines.
4. The underside, the same way as the top.
5. The other side face: the splice's center line, and the peg center lines evenly either side of it, a quarter of the splice length out.

3 · Write the cutting order on it

1. End cut; mark the end grain.
2. Peg holes first, about a quarter of the width deep, from the side face that stays whole.
3. Rip from the end to just past the slope; crosscut at the slope.
4. Crosscut the shoulder; split the next layer off from the slope end.
5. Mark the slope again; saw it; kerfs back to the shoulder; chop; pare.
6. The housing; the tongue; finish the peg holes.

Check before cutting

- Which side stays at each stage? Shade the waste.
- Does every cut leave half the line?
- Mark a second block as the upper piece: the same shape, with its slope reversed.

The sequence follows one training course (ものつくり大学 WBT, day 6). Sizes are scaled from one lab's test proportions (Toda 2012), not a carpentry standard.